CORE PURE / VECTORS, MATRICES & LINEAR TRANSFORMATIONS

Reflections in Planes

Mark scheme 7 questions · 81 marks

Questions

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	Total	**81**	

Reading the mark scheme

These rules apply throughout, unless a note says otherwise.

Marks

Every mark is worth one, so **M1 A1** on a row is two marks. Only marks coded ft follow through.

Code	Meaning
M1	Method: a correct method, applied to this question. A slip doesn't cost it if the method is clear. A formula must be used, not just quoted.
dM1	Dependent method: as M1, but only if the method mark before it was earned, unless its note names another.
A1	Accuracy: a correct answer or step. It needs the method mark it follows.
A1ft	As A1, following through from an earlier answer that the note names.
A1 AG	The last mark of a part whose answer is printed in the question: complete working, with no errors.
B1	A correct value, fact or statement. It needs no method mark.
B1ft	As B1, following through from an earlier answer that the note names.
E1	A written explanation, interpretation or conclusion. Concluding a proof by induction needs every earlier mark.

Abbreviations

ft and AG qualify a code in the Marks column. The others appear in the notes.

ft	follow through	**AG**	answer given in the question
cao	correct answer only, with no follow-through	**cso**	correct solution only: no errors in the part
oe	or an equivalent form	**awrt**	any answer that rounds to this value
soi	seen, or implied by later working	**isw**	ignore working after a correct answer
SC	special case: credit outside the main method	**their**	the student's own earlier value
condone	a slip that doesn't cost the mark	[ ]	working the mark doesn't need
s.f.	significant figures	**d.p.**	decimal places

Reading the scheme

- Each row is a line of working, with the marks it earns beside it. Where a row carries two marks, each note starts with its code.
- [$\boldsymbol{n}$] ends each part with its total.
- A shaded *Alternative* is a second method for the same total. Mark an answer by the method it follows.
- Any other valid method earns the equivalent marks, unless the question names a method or says *hence*.
- A sketch earns one B1 for each feature named below its figure.

Accuracy and answers

- Unless a question says otherwise, give answers to 3 significant figures, and angles in degrees to 1 decimal place. Angles in radians take 3 significant figures.
- A final answer with more figures is accepted if it rounds to the scheme's value; one with fewer is not. An answer spoiled by rounding too early loses its accuracy mark.
- Where a question asks for an exact answer, a decimal loses the final accuracy mark. Otherwise either is accepted.
- Wrong or missing units cost nothing unless the question asks for them.
- An omitted $\mathrm{d}\theta$, $\mathrm{d}x$ or $\mathrm{d}u$ in an integral is not penalised where the variable of integration is clear from the integrand and the limits. It is penalised where it is genuinely ambiguous: a substitution part-way through, two variables in play, or limits in the wrong variable.
- A correct answer with no working earns full marks, unless the question needs the working: show that, prove, hence, a method it names, or an exact answer. A wrong answer with no working earns nothing. Calculator answers are treated the same way.
- A correct answer, once reached, isn't spoiled by later working (isw).

Value of g

- Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question states a value, which is then the only one accepted. Answers that use $g = 9.8$ may be given to 2 or 3 significant figures.
- Where a question doesn't state g, an answer worked with $g = 10$ throughout earns the same marks, keeping any value printed in the question. The worked solutions list the final answers with $g = 10$.

Question	Scheme	Marks	Notes
1	$\overrightarrow{CD} = 2\mathbf{i} + \mathbf{j} - \mathbf{k}$, $\overrightarrow{CE} = \mathbf{i} + 3\mathbf{j}$ $\mathbf{n} = \overrightarrow{CD} \times \overrightarrow{CE} = \ldots$	**M1**	Finds two vectors in the plane from the given points and attempts their vector product, or solves $\mathbf{n} \cdot \overrightarrow{CD} = \mathbf{n} \cdot \overrightarrow{CE} = 0$. Condone sign slips.
	$\mathbf{n} = 3\mathbf{i} - \mathbf{j} + 5\mathbf{k}$	**A1**	Any nonzero multiple.
	$d = (\mathbf{i} - \mathbf{j} + 2\mathbf{k}) \cdot "(3\mathbf{i} - \mathbf{j} + 5\mathbf{k})" = \ldots$	**M1**	Forms the scalar product of their normal with the position vector of C, D or E.
	$3x - y + 5z = 14$	**A1**	oe, including $\mathbf{r} \cdot (3\mathbf{i} - \mathbf{j} + 5\mathbf{k}) = 14$ or any multiple.
	$\mathbf{p}' = \mathbf{p} - 2\dfrac{\mathbf{p} \cdot \mathbf{n} - d}{\lvert\mathbf{n}\rvert^2}\mathbf{n}$, $\mathbf{p} = 4\mathbf{i} - 2\mathbf{j} + 7\mathbf{k}$	**M1**	Uses the reflection formula with A and their plane, or forms the line through A in the direction of their normal.
	$\mathbf{p} \cdot \mathbf{n} - d = 12 + 2 + 35 - 14$ $[= 35]$ $\lvert\mathbf{n}\rvert^2 = 9 + 1 + 25$ $[= 35]$	**dM1**	Evaluates $\mathbf{p} \cdot \mathbf{n} - d$ and $\lvert\mathbf{n}\rvert^2$ with their values, or substitutes their line into their plane and solves for the parameter.
	$\mathbf{p}' = (4\mathbf{i} - 2\mathbf{j} + 7\mathbf{k})$ $\quad - 2\left(\frac{"35"}{"35"}\right)(3\mathbf{i} - \mathbf{j} + 5\mathbf{k})$	**dM1**	Subtracts twice their projection term times $\mathbf{n}$ from $\mathbf{p}$, or uses the foot F with $\overrightarrow{OB} = \overrightarrow{OA} + 2\overrightarrow{AF}$ oe.
	$B = (-2, 0, -3)$	**A1**	Accept a position vector.
	Total for Question 1	**8**	

Question	Scheme	Marks	Notes
2(i)	$\mathbf{n} = \begin{pmatrix}2\\1\\0\end{pmatrix} \times \begin{pmatrix}2\\-1\\1\end{pmatrix} = \ldots$	**M1**	Forms the vector product of the two direction vectors, or finds a vector perpendicular to both by any method. Condone sign slips.
	$\mathbf{n} = \begin{pmatrix}1\\-2\\-4\end{pmatrix}$	**A1**	Or any nonzero multiple.
	$p = \begin{pmatrix}1\\-2\\-4\end{pmatrix} \cdot \begin{pmatrix}1\\1\\2\end{pmatrix} = \ldots$	**M1**	Forms the scalar product of their normal with a point on m.
	$\mathbf{r} \cdot \begin{pmatrix}1\\-2\\-4\end{pmatrix} = -9$	**A1**	oe, the same multiple of normal and p. Must be in the form $\mathbf{r} \cdot \mathbf{n} = p$; a Cartesian equation alone scores A0.
		[4]	
2(ii)	$\dfrac{\left\lvert \begin{pmatrix}4\\2\\-3\end{pmatrix} \cdot \begin{pmatrix}1\\-2\\-4\end{pmatrix} - (-9) \right\rvert}{\sqrt{1^2 + (-2)^2 + (-4)^2}}$	**M1**	Uses the point-to-plane distance with a point on ℓ and their plane from (i), or any valid method such as the foot of the perpendicular. Condone sign slips.
	$= \sqrt{21}$	**A1**	oe, or awrt 4.58.
		[2]	
2(iii)	$\begin{pmatrix}2\\-1\\1\end{pmatrix} \cdot \begin{pmatrix}1\\-2\\-4\end{pmatrix} = 0 \Rightarrow$ direction $\begin{pmatrix}2\\-1\\1\end{pmatrix}$ unchanged	**B1**	Direction of ℓ or a multiple, justified by $\mathbf{b} \cdot \mathbf{n} = 0$ or ℓ parallel to Π, or found from two reflected points.
	$\begin{pmatrix}4\\2\\-3\end{pmatrix} - 2\dfrac{\begin{pmatrix}4\\2\\-3\end{pmatrix} \cdot \mathbf{n} - (-9)}{\lvert\mathbf{n}\rvert^2}\mathbf{n}$	**M1**	Applies the reflection formula to a point of ℓ with their normal, or forms the line through that point along their normal.
	$\begin{pmatrix}4\\2\\-3\end{pmatrix} \cdot \mathbf{n} - (-9) = 21, \quad \lvert\mathbf{n}\rvert^2 = 21$	**dM1**	Evaluates both quantities, or substitutes that line into their plane and solves for its parameter. Condone sign slips.

Question 2 continued

Question	Scheme	Marks	Notes
	$\begin{pmatrix}4\\2\\-3\end{pmatrix} - 2\begin{pmatrix}1\\-2\\-4\end{pmatrix} = \ldots\ [= (2,6,5)]$	**dM1**	Subtracts twice their component along the normal, or adds to the point twice its displacement to their foot of the perpendicular.
	$\mathbf{r} = \begin{pmatrix}2\\6\\5\end{pmatrix} + t\begin{pmatrix}2\\-1\\1\end{pmatrix}$	**A1**	oe, any point of the line with any multiple of the direction. Must be an equation $\mathbf{r} = \ldots$; the point alone scores A0.
		[5]	
	Total for Question 2	**11**	

Question	Scheme	Marks	Notes
3(a)	$3x - y + 2z = 6$ $3(1+2\lambda) - (4+\lambda) + 2\lambda = 6$ $\Rightarrow \lambda = \ldots$ $[= 1]$	**M1**	Forms a Cartesian or scalar-product equation of Π_1, substitutes the general point of ℓ_1 and solves for λ. Condone sign or arithmetic slips.
	$(1 + 2 \times "1",\ 4 + "1",\ "1")$	**dM1**	Substitutes their value of λ into the equation of ℓ_1.
	$(3, 5, 1)$	**A1**	Accept the position vector.
		[3]	
3(b)	Reflects $(1, 4, 0)$ along the normal: $\mathbf{p}' = \mathbf{p} - 2\,\frac{\mathbf{p}\cdot\mathbf{n}-6}{\lvert\mathbf{n}\rvert^2}\,\mathbf{n}$	**M1**	Uses the reflection formula with a point of ℓ_1 not on Π_1, or forms the line through it along $\mathbf{n}$.
	$\mathbf{p}\cdot\mathbf{n} - 6 = -7$, $\lvert\mathbf{n}\rvert^2 = 14$	**dM1**	Evaluates $\mathbf{p}\cdot\mathbf{n} - d$ and $\lvert\mathbf{n}\rvert^2$, or solves for the parameter where the line along $\mathbf{n}$ meets Π_1. Condone arithmetic slips.
	$\mathbf{p}' = \begin{pmatrix}1\\4\\0\end{pmatrix} - 2\left(\frac{"-7"}{"14"}\right)\begin{pmatrix}3\\-1\\2\end{pmatrix}$	**dM1**	Subtracts twice their projection term times $\mathbf{n}$, or adds twice the displacement from the point to their foot of the perpendicular.
	$= \begin{pmatrix}4\\3\\2\end{pmatrix}$	**A1**	Correct image, with no incorrect values seen.
	$\mathbf{b}' = \begin{pmatrix}2\\1\\1\end{pmatrix} - 2\left(\frac{7}{14}\right)\begin{pmatrix}3\\-1\\2\end{pmatrix}$	**M1**	Finds the reflected direction of ℓ_1 by the formula with no d term, or as the vector joining their $(3, 5, 1)$ from (a) to their image.
	$= \begin{pmatrix}-1\\2\\-1\end{pmatrix}$ $\mathbf{r} = \begin{pmatrix}4\\3\\2\end{pmatrix} + \mu\begin{pmatrix}-1\\2\\-1\end{pmatrix}$	**A1 AG**	cso. Correct image and a direction parallel to $(-1, 2, -1)$ reached without error, then the printed equation, showing their image lies on the printed line, or $(4, 3, 2)$ on theirs, if another point was reflected.
	Alternative: Fixed point from (a)		
	$P(3, 5, 1)$ from (a) lies in Π_1, so is on ℓ_2	**B1**	Stated, or used as a point of ℓ_2.
	$\mathbf{b}' = \mathbf{b} - 2\,\frac{\mathbf{b}\cdot\mathbf{n}}{\lvert\mathbf{n}\rvert^2}\,\mathbf{n}$	**M1**	Finds the reflected direction of ℓ_1 by the formula with no d term, or reflects a second point of ℓ_1 in Π_1.

Question 3 continued

Question	Scheme	Marks	Notes
	$= \begin{pmatrix}2\\1\\1\end{pmatrix} - 2\left(\frac{7}{14}\right)\begin{pmatrix}3\\-1\\2\end{pmatrix}$	**dM1**	Evaluates $\mathbf{b}\cdot\mathbf{n}$ and $\lvert\mathbf{n}\rvert^2$ and subtracts twice the projection, or joins their image point to their point from (a). Condone arithmetic slips.
	$= \begin{pmatrix}-1\\2\\-1\end{pmatrix}$	**A1**	Or any nonzero multiple, with no incorrect values seen.
	$\mu = 1$: $\begin{pmatrix}4\\3\\2\end{pmatrix} + \begin{pmatrix}-1\\2\\-1\end{pmatrix} = \begin{pmatrix}3\\5\\1\end{pmatrix}$	**M1**	Substitutes their $(3,5,1)$ into the printed equation, finding μ and checking all three components; or checks $(4,3,2)$ on their line through it.
	Same point and same direction, so ℓ_2 is the printed line	**A1 AG**	cso. Direction parallel to $(-1,2,-1)$, and $(3,5,1)$ on the printed line or $(4,3,2)$ on theirs, without error, with a conclusion.
		[6]	
3(c)	$\mathbf{b}+\mathbf{b}' = \begin{pmatrix}2\\1\\1\end{pmatrix} + \begin{pmatrix}-1\\2\\-1\end{pmatrix} = \ldots$ through $(3,5,1)$	**M1**	Finds a direction in both planes, such as $\mathbf{b}+\mathbf{b}'$ or the cross product of the planes' normals, and uses it with a point on both, such as their $(3,5,1)$.
	$\mathbf{r} = \begin{pmatrix}3\\5\\1\end{pmatrix} + t\begin{pmatrix}1\\3\\0\end{pmatrix}$	**A1**	oe, with any point on the line and any nonzero multiple of $(1,3,0)$.
		[2]	
	Total for Question 3	**11**	

Question	Scheme	Marks	Notes
4(a)	$\overrightarrow{BA} = -\mathbf{i} - 2\mathbf{j} - \mathbf{k},\ \overrightarrow{BC} = -\mathbf{i} - \mathbf{j} + (c-1)\mathbf{k}$ $\overrightarrow{BA} \times \overrightarrow{BC} = \ldots$	**M1**	Forms both vectors and expands their vector product by determinant or components. Condone sign slips in the vectors.
	$(1-2c)\mathbf{i} + c\mathbf{j} - \mathbf{k}$	**A1**	oe, column vector accepted.
		[2]	
4(b)	$(1-2c)\mathbf{i} + c\mathbf{j} - \mathbf{k} = d\mathbf{i} + 2\mathbf{j} - \mathbf{k}$ $\Rightarrow c = \ldots,\ d = 1 - 2c$	**M1**	Equates corresponding components of their vector from (a), in terms of c, and the given vector to find c.
	$c = 2,\ d = 1 - 2(2) = -3$	**A1 AG**	$c = 2$ and $d = -3$ from a correct (a), with $c = 2$ substituted into $d = 1 - 2c$.
		[2]	
4(c)	$\mathbf{n} = -3\mathbf{i} + 2\mathbf{j} - \mathbf{k}$ $p = \mathbf{i} \cdot (-3\mathbf{i} + 2\mathbf{j} - \mathbf{k}) = \ldots$	**M1**	Takes a normal parallel to $-3\mathbf{i} + 2\mathbf{j} - \mathbf{k}$ and forms its scalar product with the position vector of A, B or C, using $c = 2$ for C.
	$\mathbf{r} \cdot (-3\mathbf{i} + 2\mathbf{j} - \mathbf{k}) = -3$	**A1**	Or any multiple, such as $\mathbf{r} \cdot (3\mathbf{i} - 2\mathbf{j} + \mathbf{k}) = 3$. A Cartesian equation alone scores A0.
		[2]	
4(d)	$\mathbf{s}' = \mathbf{s} - 2\,\frac{\mathbf{s}\cdot\mathbf{n} - p}{\lvert\mathbf{n}\rvert^2}\,\mathbf{n}$	**M1**	Uses a correct reflection formula with $\mathbf{s} = 3\mathbf{i} + \mathbf{j} + 2\mathbf{k}$ and their $\mathbf{n}$ and p from (c). If not quoted, implied by correct working.
	$\mathbf{s} \cdot \mathbf{n} - p = -9 - (-3) = -6$ $\lvert\mathbf{n}\rvert^2 = 14$	**dM1**	Evaluates both $\mathbf{s} \cdot \mathbf{n} - p$ and $\lvert\mathbf{n}\rvert^2$ for their $\mathbf{n}$ and p. Condone arithmetic slips.
	$\mathbf{s}' = \mathbf{s} - 2\left(\text{"}-\frac{6}{14}\text{"}\right)\mathbf{n}$ $= (3\mathbf{i} + \mathbf{j} + 2\mathbf{k}) + \frac{6}{7}(-3\mathbf{i} + 2\mathbf{j} - \mathbf{k})$	**dM1**	Subtracts from $\mathbf{s}$ twice their scalar $\frac{\mathbf{s}\cdot\mathbf{n} - p}{\lvert\mathbf{n}\rvert^2}$ times their $\mathbf{n}$. Need not be simplified.
	$\frac{3}{7}\mathbf{i} + \frac{19}{7}\mathbf{j} + \frac{8}{7}\mathbf{k}$	**A1**	oe, or awrt $0.429\mathbf{i} + 2.71\mathbf{j} + 1.14\mathbf{k}$. Column vector accepted.
	Alternative: Line through S normal to Π		
	$\mathbf{r} = (3\mathbf{i} + \mathbf{j} + 2\mathbf{k}) + \lambda(-3\mathbf{i} + 2\mathbf{j} - \mathbf{k})$	**M1**	Forms the line through S with direction their normal from (c).

Question 4 continued

Question	Scheme	Marks	Notes
	$-3(3-3\lambda)+2(1+2\lambda)-(2-\lambda)=-3$ $\Rightarrow \lambda = \ldots$ $[= \frac{3}{7}]$	**dM1**	Substitutes the general point of their line into their plane equation and solves for λ. Condone sign slips.
	$\mathbf{s}' = \mathbf{s} + 2 \times$ "$\frac{3}{7}$" $\mathbf{n}$ $= (3\mathbf{i}+\mathbf{j}+2\mathbf{k}) + \frac{6}{7}(-3\mathbf{i}+2\mathbf{j}-\mathbf{k})$	**dM1**	Uses the foot of the perpendicular as the midpoint of SS', by doubling their λ or by $\mathbf{s}' = 2\mathbf{m} - \mathbf{s}$ with $\mathbf{m}$ the foot.
	$\frac{3}{7}\mathbf{i} + \frac{19}{7}\mathbf{j} + \frac{8}{7}\mathbf{k}$	**A1**	oe, or awrt $0.429\mathbf{i} + 2.71\mathbf{j} + 1.14\mathbf{k}$. Column vector accepted.
		[4]	
Total for Question 4		**10**	

Question	Scheme	Marks	Notes
5(a)	$2(1+\lambda)-2\lambda+(1+3\lambda)=9$ $\Rightarrow \lambda = \ldots \ [=2]$	**M1**	Substitutes the general point of ℓ into the equation of Π, in either form, and solves for λ. Condone sign slips.
	$\mathbf{r}=(\mathbf{i}+\mathbf{k})+"2"(\mathbf{i}+2\mathbf{j}+3\mathbf{k})$	**dM1**	Substitutes their λ into the equation of ℓ.
	$(3,4,7)$	**A1**	Accept a position vector.
		[3]	
5(b)	$\mathbf{r}=4\mathbf{j}+\mathbf{k}+t(2\mathbf{i}-\mathbf{j}+\mathbf{k})$	**M1**	Forms the line through A along $2\mathbf{i}-\mathbf{j}+\mathbf{k}$, in any form, or finds $\overrightarrow{AB}$ and compares it with the normal.
	$2(2t)-(4-t)+(1+t)=9 \Rightarrow t=\ldots$	**dM1**	Substitutes their general point into the equation of Π and solves for the parameter, or checks $(4,2,3)$ in $2x-y+z=9$.
	$t=2 \Rightarrow B=(2\times 2,\ 4-2,\ 1+2)$ $=(4,2,3)$	**A1 AG**	Correct substituted equation leading without error to $t=2$ and $(4,2,3)$; or both checks correct, with $\overrightarrow{AB}=2(2\mathbf{i}-\mathbf{j}+\mathbf{k})$ and a conclusion.
		[3]	
5(c)	$\mathbf{p}'=(4\mathbf{j}+\mathbf{k})-2\frac{-3-9}{6}(2\mathbf{i}-\mathbf{j}+\mathbf{k})$	**M1**	Uses $\mathbf{p}-2\frac{\mathbf{p}\cdot\mathbf{n}-d}{\lvert\mathbf{n}\rvert^2}\mathbf{n}$ with $\mathbf{n}=2\mathbf{i}-\mathbf{j}+\mathbf{k}$ and $d=9$. Condone slips if quoted; if not, implied by correct values.
	$A'=(8,0,5)$	**A1**	Accept a position vector.
	Alternative: Using B as the midpoint		
	$\overrightarrow{OA'}=\overrightarrow{OA}+2\overrightarrow{AB}$ $=4\mathbf{j}+\mathbf{k}+2(4\mathbf{i}-2\mathbf{j}+2\mathbf{k})$	**M1**	Uses B as the midpoint of AA', by $\overrightarrow{OA}+2\overrightarrow{AB}$, $2\overrightarrow{OB}-\overrightarrow{OA}$ or $t=4$ in the line from (b).
	$A'=(8,0,5)$	**A1**	Accept a position vector.
		[2]	
5(d)	$(3,4,7)$ lies in Π, so is on the reflected line	**B1**	Stated, or used as a point of the reflected line.
	Reflects the direction of ℓ: $\mathbf{u}'=\mathbf{u}-2\frac{\mathbf{u}\cdot\mathbf{n}}{\lvert\mathbf{n}\rvert^2}\mathbf{n}$	**M1**	Finds the reflection of $\mathbf{u}=\mathbf{i}+2\mathbf{j}+3\mathbf{k}$ using $\mathbf{n}$, by the formula with no constant term, or reflects a second point of ℓ in Π.

Question 5 continued

Question	Scheme	Marks	Notes
	$\mathbf{u}' = (\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}) - 2\left(\frac{3}{6}\right)(2\mathbf{i} - \mathbf{j} + \mathbf{k})$	**dM1**	Evaluates $\mathbf{u} \cdot \mathbf{n}$ and $\lvert\mathbf{n}\rvert^2$ and subtracts twice the projection, or joins the image point to their point from (a). Condone arithmetic slips.
	$\mathbf{u}' = -\mathbf{i} + 3\mathbf{j} + 2\mathbf{k}$	**A1**	Or any nonzero multiple.
	$\mathbf{r} = 3\mathbf{i} + 4\mathbf{j} + 7\mathbf{k} + \mu(-\mathbf{i} + 3\mathbf{j} + 2\mathbf{k})$	**A1ft**	ft their point from (a) and their direction; soi by their $\mathbf{r} \times \mathbf{a} = \mathbf{b}$.
	$\mathbf{r} \times \mathbf{a} = \mathbf{c} \times \mathbf{a}$, $\mathbf{c} = 3\mathbf{i} + 4\mathbf{j} + 7\mathbf{k}$, $\mathbf{a} =$ "$-\mathbf{i} + 3\mathbf{j} + 2\mathbf{k}$"	**M1**	Takes $\mathbf{a}$ as their direction and $\mathbf{b}$ as the cross product of a point of their line with it. Condone the order of the product.
	$(3\mathbf{i} + 4\mathbf{j} + 7\mathbf{k}) \times (-\mathbf{i} + 3\mathbf{j} + 2\mathbf{k})$ $= (8 - 21)\mathbf{i} - (6 + 7)\mathbf{j} + (9 + 4)\mathbf{k}$	**dM1**	Attempts the cross product of their point and direction, by determinant or components. Condone sign slips.
	$\mathbf{r} \times (-\mathbf{i} + 3\mathbf{j} + 2\mathbf{k}) = -13\mathbf{i} - 13\mathbf{j} + 13\mathbf{k}$	**A1**	Or with $\mathbf{a}$ and $\mathbf{b}$ both multiplied by the same nonzero k. Must be in the form $\mathbf{r} \times \mathbf{a} = \mathbf{b}$.
		[8]	
Total for Question 5		**16**	

Question	Scheme	Marks	Notes
6(a)	$\begin{pmatrix}2\\-1\\1\end{pmatrix}\times\begin{pmatrix}1\\1\\0\end{pmatrix}=\ldots$	**M1**	Attempts the vector product of the two direction vectors, or solves two scalar products equal to zero, for a normal.
	$\mathbf{n}=\begin{pmatrix}-1\\1\\3\end{pmatrix}$	**A1**	Or any nonzero multiple.
	$-x+y+3z=-2$	**B1**	oe, in vector or Cartesian form.
	$-(4-\lambda)+(1+\lambda)+3(2+3\lambda)=-2$ $\Rightarrow\lambda=\ldots$	**M1**	Substitutes the general point of a line through P along their normal into their plane equation and solves for the parameter.
	$\lambda=-\frac{5}{11}$	**A1**	Or the correct value for a multiple of $\mathbf{n}$.
	$\overrightarrow{OQ}=\begin{pmatrix}4\\1\\2\end{pmatrix}-\frac{5}{11}\begin{pmatrix}-1\\1\\3\end{pmatrix}$	**dM1**	Substitutes their λ into their line through P.
	$Q\left(\frac{49}{11},\frac{6}{11},\frac{7}{11}\right)$	**A1**	Or awrt $(4.45, 0.545, 0.636)$. A position vector is accepted.
		[7]	
6(b)	$\overrightarrow{PQ}=\frac{5}{11}\begin{pmatrix}1\\-1\\-3\end{pmatrix}$, $\overrightarrow{QR}=\frac{1}{11}\begin{pmatrix}-16\\5\\-7\end{pmatrix}$ $\begin{pmatrix}1\\-1\\-3\end{pmatrix}\times\begin{pmatrix}-16\\5\\-7\end{pmatrix}=\ldots$	**M1**	Forms two non-parallel vectors in the plane from P, R, their Q or $\mathbf{n}$, and attempts their vector product.
	$\begin{pmatrix}22\\55\\-11\end{pmatrix}=11\begin{pmatrix}2\\5\\-1\end{pmatrix}$	**A1**	Or any nonzero multiple.
	$2x+5y-z=2(4)+5(1)-1(2)$	**M1**	Uses their normal with P, R or their Q to find the constant, in vector or Cartesian form.
	$2x+5y-z=11$	**A1**	oe Cartesian equation. A vector form alone scores A0.
		[4]	
6(c)	R lies on Π, so is fixed: R is on ℓ_3	**B1**	soi by using R as a point of ℓ_3.
	Reflects the direction of ℓ_2 in Π: $\begin{pmatrix}1\\0\\2\end{pmatrix}-2\times\frac{5}{11}\begin{pmatrix}-1\\1\\3\end{pmatrix}=\ldots$	**M1**	Uses $\mathbf{b}-2\frac{\mathbf{b}\cdot\mathbf{n}}{\lvert\mathbf{n}\rvert^2}\mathbf{n}$ with their normal, or reflects P in Π, such as by $\overrightarrow{OP'}=2\overrightarrow{OQ}-\overrightarrow{OP}$ with their Q, and finds $\overrightarrow{RP'}$.

Question 6 continued

Question	Scheme	Marks	Notes
	$\frac{1}{11}\begin{pmatrix}21\\-10\\-8\end{pmatrix}$	**A1**	Or any nonzero multiple.
	$\mathbf{r}=\begin{pmatrix}3\\1\\0\end{pmatrix}+\mu\begin{pmatrix}21\\-10\\-8\end{pmatrix}$	**A1ft**	ft their direction, through R or their P'; any nonzero multiple. Needs $\mathbf{r}=\ldots$ with a parameter.
		[4]	
	Total for Question 6	**15**	

Question	Scheme	Marks	Notes
7	$\begin{pmatrix}1\\1\\1\end{pmatrix} \times \begin{pmatrix}2\\-1\\1\end{pmatrix} = \begin{pmatrix}2\\1\\-3\end{pmatrix}$	**M1 A1**	**M1** Attempts the vector product of the two normals, or the difference of two points on ℓ_1. **A1** Or any nonzero multiple.
	$x+y+z=1,\ 2x-y+z=4$ $x+y+2z=3$	**M1**	Uses the equation of Π with both plane equations, or substitutes their parametric form of ℓ_1 into Π.
	$\Rightarrow (x,y,z) = \ldots$	**dM1**	Solves by any method, calculator allowed, to a point.
	$\left(\frac{1}{3}, -\frac{4}{3}, 2\right)$	**A1**	
	The point lies on Π, so is on ℓ_2	**B1**	Stated, or implied by its use as the point on ℓ_2.
	$\mathbf{u}' = \mathbf{u} - 2\,\frac{\mathbf{u}\cdot\mathbf{n}}{\lvert\mathbf{n}\rvert^2}\,\mathbf{n}$	**M1**	Finds the reflection of their direction in Π by the formula with no d term; or reflects a second point of ℓ_1 in Π.
	$= \text{"}\begin{pmatrix}2\\1\\-3\end{pmatrix}\text{"} - 2\left(\frac{-3}{6}\right)\begin{pmatrix}1\\1\\2\end{pmatrix}$	**dM1**	Finds $\mathbf{u}\cdot\mathbf{n}$ and $\lvert\mathbf{n}\rvert^2$ and subtracts twice the projection; or subtracts the fixed point from the image point. Condone arithmetic slips.
	$= \begin{pmatrix}3\\2\\-1\end{pmatrix}$	**A1**	Or any nonzero multiple.
	$\mathbf{r} = \text{"}\begin{pmatrix}\frac{1}{3}\\-\frac{4}{3}\\2\end{pmatrix}\text{"} + t\text{"}\begin{pmatrix}3\\2\\-1\end{pmatrix}\text{"}$	**A1ft**	ft their point and reflected direction; any point of ℓ_2, such as the image of their second point, and any nonzero multiple. Needs $\mathbf{r} = \ldots$ with a parameter.
	Total for Question 7	**10**	